Physics Study Guide

RC Circuits & Square Wave Signals

This study guide breaks down the behavior of an RC (Resistor-Capacitor) circuit when excited by a Low-Frequency Generator (LFG) providing a square wave signal.

1. The Experimental Setup

When analyzing RC circuits in the lab or on an exam, you are often asked about the oscilloscope setup.

- **The Circuit:** An LFG, a capacitor (C), and a resistor (R) in series.
- **Oscilloscope Connections:**
  - Channel 1 (Y_1): Connected across the capacitor to measure u_C (u_{AM}).
  - Channel 2 (Y_2): Connected across the resistor to measure $-u_R$ (u_{BM}).

Exam Pro-Tip

Because Channel 2 measures the voltage with respect to the common ground (Mass), it reads $-u_R$. To correctly visualize u_R (which is u_{MB}) on the screen, you must press the **INVERT** (or REV) button on Channel 2.

2. The Square Wave Signal & Phases

The LFG outputs a square voltage u_G with a period T. This splits the circuit's behavior into two distinct phases:

1. **Phase 1: The Charging Phase ($0 \leq t \leq \frac{T}{2}$)**
   - Voltage: $u_G = E$ (Constant max voltage).
   - Current (i): Flows in the positive direction ($i > 0$).
   - Charge (q): The capacitor gains charge over time ($\frac{dq}{dt} > 0$).
2. **Phase 2: The Discharging Phase ($\frac{T}{2} \leq t \leq T$)**
   - Voltage: $u_G = 0$ (Generator acts like a closed wire).
   - Current (i): Flows in the opposite direction ($i < 0$).
   - Charge (q): The capacitor loses charge ($\frac{dq}{dt} < 0$).

The Golden Rule: In both phases, the relationship between current and charge remains $i = \frac{dq}{dt}$.

□ 3. The Time Constant (τ) & Complete Cycles

The time constant $\tau = RC$ dictates how fast the capacitor charges or discharges.

- **Rule of Thumb:** A capacitor is considered fully charged (or fully discharged) after 5τ.
- **Condition for a Complete Cycle:** For the capacitor to fully charge and fully discharge within one period T of the LFG, the half-period must be at least 5τ:

$$5\tau \leq \frac{T}{2} \implies T \geq 10\tau \implies T \geq 10RC$$

□ 4. Mathematical Master Sheet

You will often be asked to derive or use the differential equations and their temporal solutions. Use the Law of Addition of Voltages ($u_G = u_c + u_R$) as your starting point.

Variable	Charging Phase ($u_G = E$)	Discharging Phase ($u_G = 0$)
Diff. Eq. for u_C	$\frac{du_c}{dt} + \frac{u_c}{RC} = \frac{E}{RC}$	$\frac{du_c}{dt} + \frac{u_c}{RC} = 0$
Solution for $u_C(t)$	$u_c(t) = E(1 - e^{-\frac{t}{RC}})$	$u_c(t) = E \cdot e^{-\frac{t}{RC}}$
Solution for $q(t)$	$q(t) = C \cdot E(1 - e^{-\frac{t}{RC}})$	$q(t) = C \cdot E \cdot e^{-\frac{t}{RC}}$
Solution for $i(t)$	$i(t) = \frac{E}{R}e^{-\frac{t}{RC}}$	$i(t) = -\frac{E}{R}e^{-\frac{t}{RC}}$
Solution for $u_R(t)$	$u_R(t) = E \cdot e^{-\frac{t}{RC}}$	$u_R(t) = -E \cdot e^{-\frac{t}{RC}}$

□ 5. Graphical Analysis (What the curves look like)

Capacitor Voltage (u_C):

- **Charge:** Starts at 0, curves upward, and flattens out at an asymptote of E.
- **Discharge:** Starts at E, curves downward, and flattens out at 0.
- **How to find τ graphically:** Draw a tangent line at $t = 0$. Where this tangent intersects the horizontal asymptote (E for charge, 0 for discharge), drop a vertical line to the time axis. That time is τ. Alternatively, find the time when $u_C = 0.63E$ (charge) or $u_C = 0.37E$ (discharge).

Resistor Voltage (u_R) & Current (i):

- Both follow the exact same shape because $u_R = R \cdot i$.
- **Charge:** Spikes instantly to maximum (E for u_R, $\frac{E}{R}$ for i) at $t = 0$, then decays exponentially to 0.
- **Discharge:** Plummets instantly to a negative maximum ($-E$ for u_R, $-\frac{E}{R}$ for i) at $t = \frac{T}{2}$, then decays upwards back to 0.

6. Common Pitfalls to Avoid

- **Forgetting the negative sign during discharge:** Remember that during discharge, current flows backwards. Therefore, i and u_R must be negative!
- **Confusing the Differential Equations:** The diff eq for u_C equals $\frac{E}{RC}$ during charge, but the diff eq for i and u_R always equals 0 on the right side, regardless of the phase.
- **Applying $q = C \cdot u_C$ incorrectly:** This rule is always true. If you know the behavior of u_C, you know the behavior of q (just multiply by C).

PHYSICS DEPARTMENT
Rabih A.G.